

19: Binary Logistic Regression

Stat 230 - Spring 2026

1 Fitting a Binary Logistic Regression Model

Fitting a binary logistic regression model in R is similar to fitting a linear regression model. To guide you through this process, let's revisit the fake news dataset from class.

To load the data, run the following code chunk:

```
# news <- read.csv("https://aluby.github.io/stat230/data/fake_news_bayesrules.csv")
news <- read.csv(here::here("data/fake_news_bayesrules.csv"))
```

Binary logistic regression requires a binary response variable. In this case, we will use the `type_num` variable, which indicates whether the article was labeled as fake (1 = yes, 0 = no). Be sure to always check that your response variable is coded correctly before modeling. In this case, we want to make sure that `type` and `type_num` match up with our understanding of how the numeric variable is coded.

Next, we will fit a binary logistic regression model using the `glm()` function in R. The syntax is similar to that of the `lm()` function, but we need to specify the `family` argument as `binomial` to indicate that we are performing logistic regression. Here, we will model the probability of an article being fake news by the number of words in the title.

```
news_glm <- glm(type_num ~ title_words, data = news, family = binomial)
```

You can still use the `summary()` function to view the results of the model fit:

```
summary(news_glm)
```

Call:

```
glm(formula = type_num ~ title_words, family = binomial, data = news)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.2058	0.6090	-3.622	0.000292	***
title_words	0.1588	0.0510	3.114	0.001844	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 201.90 on 149 degrees of freedom
Residual deviance: 191.35 on 148 degrees of freedom
AIC: 195.35

Number of Fisher Scoring iterations: 4

Much of the regression output looks the same as in linear regression, but there are some key differences. The main differences are:

- The test statistic is now labeled z value instead of t value.
- A Null deviance and Residual deviance are reported instead of Multiple R-squared and Adjusted R-squared.

2 Making predictions

We can also use R to calculate the predicted probabilities from our fitted logistic regression model using the `predict()` function. Below is an example where we are calculating the predicted probability that an article that has 22 words in the title is fake news

```
predict(news_glm, newdata = data.frame(title_words = 22), type = "response")
```

```
1  
0.783819
```

Notice that we are specifying the data (x values) use for prediction using the same argument as in linear regression: we create a data frame with columns named identically to the actual data set used to fit the mode. We also must specify `type = "response"` to get probabilities. If you omit this argument, you will get a prediction on the log-odds scale, which isn't a probability!

```
predict(news_glm, newdata = data.frame(title_words = 22))
```

```
1  
1.288062
```

3 Your turn

Now, fit a logistic regression model using all three variables we've discussed today (`title_has_excl`, `title_caps_percent`, and `title_words`) and report the `summary()`

```
news_glm_2 <- glm(type_num ~ ___ + ___ + ___, data = news, family = binomial)  
summary(___)
```

- (1) Write out the fitted model (on the log-odds scale) for articles (A) with an exclamation in the title and (B) without an exclamation in the title
 - (2) Write out interpretations for all slope coefficients in the model in terms of the **odds** of being fake news
 - (3) Based on what we know about OLS (**o**rdinary **l**east **s**quares, which includes both SLR and MLR), which predictors appear to be most important for this model?
 - (4) What questions do you have about the `summary()` output
-

4 Function quick reference

The following table summarizes the functions discussed above:

Function	Purpose
<code>glm(formula, data, family = binomial)</code>	Fit a binary logistic regression model
<code>summary(model)</code>	View model summary and Wald tests
<code>predict(model, newdata, type = "response")</code>	Make predictions on the probability scale